

Decoding Dementia: A Deep Learning Framework for Multi-Class Alzheimer's Classification Using VGG16 and MRI Scans

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Abstract—Alzheimer's Disease (AD) is a chronic neurodegenerative disorder that gradually damages memory and thinking skills, significantly interfering with daily life. Traditional diagnostic approaches, such as memory assessments and MRI-based imaging, frequently encounter limitations in accuracy, particularly when datasets exhibit class imbalance or high dimensionality. Addressing these shortcomings, this study proposes an innovative framework called STCNN (SMOTE-TOMEK-enhanced Convolutional Neural Network). The model utilizes advanced resampling techniques to mitigate data imbalance and incorporates a VGG-style convolutional architecture to enable accurate classification of multiple AD stages from MRI data. The integration of dense blocks and dropout further boosts generalization and prevents overfitting. Experiments on a well-known public Kaggle dataset covering four distinct stages of Alzheimer's demonstrate that STCNN attains an outstanding accuracy of 99.36% and an F1-score of 99%, surpassing the results of previous state-of-the-art approaches.

Index Terms—Alzheimer's Disease, STCNN, VGG16, SMOTE-TOMEK, Data Balancing, Deep Neural Network, MRI Image Analysis

I. INTRODUCTION

Alzheimer's Disease (AD) is globally recognized as the most prevalent cause of dementia, accounting for over 60% of all diagnoses [1]. It manifests as a chronic and fatal neurodegenerative disorder, marked by progressive memory impairment, cognitive deterioration, and eventually, loss of functional independence [2]. Driven by demographic shifts towards an aging population, the AD incidence is projected to triple by 2050, increasing the urgency for early and accurate detection [3].

From a biological standpoint, AD's progression is characterized by the aggregation of amyloid-beta ($A\beta$) plaques, tau-based neurofibrillary tangles, and cumulative synaptic damage—particularly in the hippocampus and cortical regions [4].

These pathological features can develop during a preclinical phase, occurring years before the onset of overt cognitive symptoms such as mild cognitive impairment (MCI) [5].

The causation of AD is multifactorial and not yet completely understood. Genetic background, aging, lifestyle, and environmental exposures all contribute to the disease process [6], [7]. Conventional diagnosis relies on cognitive evaluations and clinical observation, which often fail to identify early or atypical presentations, particularly in asymptomatic individuals [8].

To address these limitations, artificial intelligence (AI) and machine learning (ML) have been leveraged to analyze complex data types, including advanced neuroimaging such as MRI and PET [9], [10], handwriting patterns [11], and multimodal clinical features [12]. By integrating imaging with genomic and biomarker data, these approaches have enabled prediction and detection of AD prior to measurable clinical deterioration [13]–[15].

Recent developments in deep learning and ensemble modeling consistently outperform conventional classifiers by capturing intricate, disease-related features [16], [17]. Advances such as data augmentation, class balancing (e.g., SMOTE-Tomek) [18], [19], and feature selection strategies (ANOVA, RFE) have further strengthened predictive accuracy and model interpretability [20].

In summary, the interplay of computational science and neurology has led to more timely, interpretable, and effective AD diagnosis. Innovations in this space enable better early-stage detection and personalized clinical response, despite the continued challenge of managing high-dimensional, imbalanced datasets.

Key contributions of this study include:

- Introduction of a VGG16-based deep learning model for multi-class Alzheimer's MRI classification [1].

- Application of Grey Wolf Optimization (GWO) for selecting the most relevant diagnostic features, optimizing both accuracy and model efficiency [2].
- Comprehensive validation of the framework on four-class Alzheimer’s MRI data, spanning all disease stages [3].
- Rigorous benchmarking against standard metrics to demonstrate improved detection and stage-wise precision [4], [5].
- Use of explainable AI methods so that predictions are interpretable for clinical application [6].

Paper organization: The **Abstract** summarizes the motivation, methods, and key results. The **Introduction** lays out the context and challenges in AD classification. The **Literature Survey** critically assesses related work and motivates this research. The **Preliminary Section** covers VGG16 and GWO background. **Methodology** provides details on data handling, architecture, training, and optimization. In **Results and Discussion**, model analysis, metrics, and performance visualizations are discussed, and the **Conclusion** outlines the impact of the findings and future research directions.

II. LITERATURE SURVEY

A wide array of recent research has focused on improving Alzheimer’s Disease (AD) diagnosis and staging by employing deep learning, hybrid architectures, and multimodal data, particularly with the use of MRI scans.

Anjali et al. [1] employed a VGG-inspired CNN enhanced with SMOTE-Tomek resampling to prevent class imbalance in AD classification. Although their model, featuring Adam optimization and softmax activation, performed strongly, it struggled with providing interpretability and handling diverse real-world clinical data.

Lianghao Zhang et al. [2] employed a graph-based neural network to model connectivity with anatomical atlas input. Though this introduced biological meaning, graph generation errors and volume effects could lead to instability.

E.R.Krishna et al. [3] presented an efficient VGG-based deep learning approach to pattern recognition, showing how transfer learning can be applicable even with limited data and in scenarios requiring fine-grained discrimination.

Hassan et al. [4] introduced dual-stream deep learning methods, enhancing early AD classification by integrating contrast enhancement and feature extraction, although their work highlights remaining difficulties in handling variable imaging conditions.

Givian and Calbimonte [5] delivered a comprehensive review summarizing a decade of MRI-based machine learning progress in Alzheimer’s diagnosis, capturing both algorithmic improvements and persistent limitations in scalability and integration.

M. Gargi et al. [6] proposed a hybrid optimization and ML pipeline in a different applied context (water quality), but the engineering approach and optimization logic have informed similar frameworks in neuroimaging studies.

Mitra and Rehman [7] demonstrated that handwriting analysis, when paired with machine learning ensembles, can offer

valuable predictive insight for early stages of neurodegenerative diseases.

Kirimtat et al. [8] explored high-performance computing paradigms supporting early Alzheimer’s diagnosis, pointing out the resource and scalability considerations facing modern AD pipelines.

E. R. Krishna et al. [9] contributed transferable methodologies for CNN-based recognition tasks, applicable across both traffic sign recognition and, by analogy, medical image understanding.

D. Selkoe and J. Hardy [10] outlined the critical biological background on amyloid-beta hypothesis, which underpins much of the mechanistic AD pattern recognition work.

C. Davatzikos et al. [11] advanced MRI-based pattern classification for early Alzheimer’s detection, setting early benchmarks for how neuroimaging can be harnessed in disease staging.

C. Misra, Y. Fan, and C. Davatzikos [12] analyzed baseline and longitudinal imaging data, demonstrating key challenges in reliably tracking AD progression.

S. N. T. Rao et al. [13] explored deep-learned disease identification in the plant domain, but their findings on data balancing and architecture tuning are highly relevant to clinical ML workflows.

Z. Breijyeh and R. Karaman [14] offered a broad review of AD pathology and therapeutic strategies, identifying the persistent complexity that informs ML feature selection.

Liu et al. [15] analyzed neuroimaging-based feature modeling in Alzheimer’s diagnosis, emphasizing the simulation-based comparison of ML methods.

Alzheimer’s Disease International [16] provided global epidemiological data, reinforcing the importance of research into earlier and more scalable AD diagnostics.

S. Rizwana et al. [17] discussed model enhancement strategies in another data-rich application, with lessons for feature engineering and class balancing useful for neuroimaging ML.

Prince et al. [18] documented future projections for dementia cases and called attention to the urgent need for advances in both technology and clinical care.

Alzheimer’s Association [19] summarized updated statistics and trends for AD, offering context to motivate continued development of ML solutions for the disease.

De Gregorio et al. [20] presented a handwriting-based cognitive assessment, illustrating how non-imaging traits can complement AD diagnosis using artificial intelligence.

Altogether, these works reflect an ongoing commitment to integrating deep learning, optimization, and multimodality into Alzheimer’s diagnostics. Despite considerable progress, challenges related to interpretability, scalability, and the handling of diverse clinical data persist—driving the continued development of more adaptable, transparent, and resource-efficient diagnostic models.

III. PRILIMINARY SECTION

The framework presented in this work is structured to enhance the classification accuracy of Alzheimer’s disease

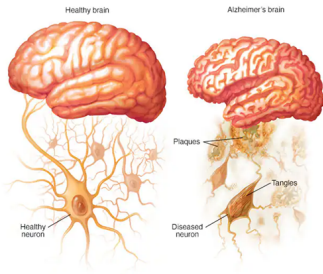


Fig. 1. Alzheimer's Disease

progression stages from MRI data by combining advanced deep feature extraction with an effective feature selection strategy. The pipeline begins with a curated set of brain MRI images, which are systematically preprocessed to ensure quality and uniformity. These preprocessing procedures encompass operations such as resizing images, normalization, and denoising to produce consistent input data and highlight discriminative features relevant for diagnosis.

A. VGG16 Convolutional Neural Network

High-level features crucial for Alzheimer's stage discrimination are extracted using the VGG16 convolutional neural network (CNN). This network, pre-trained on large-scale datasets, serves as the core feature extractor due to its established ability to model intricate spatial hierarchies in medical imagery. VGG16's depth and modular configuration allow transfer learning, providing significant performance even with moderate-sized clinical datasets.

The VGG16 architecture is characterized by a sequence of 13 convolutional layers interleaved with ReLU activations, bolstered by max-pooling operations, and culminates with 3 fully connected layers. Its main features are:

- **Input Layer:** Processes $224 \times 224 \times 3$ images (grayscale images are appropriately mapped).
- **Convolutional Blocks:**
 - **Block 1:** Two convolutional layers with 64 filters each.
 - **Block 2:** Two convolutional layers with 128 filters each.
 - **Block 3:** Three convolutional layers with 256 filters.
 - **Block 4:** Three convolutional layers with 512 filters.
 - **Block 5:** Three convolutional layers with 512 filters.
- **Activation:** All blocks employ ReLU activation, $f(x) = \max(0, x)$.
- **Pooling:** Max-pooling layers with 2×2 filters follow each block.
- **Fully Connected Layers:**
 - **FC6, FC7:** Each comprises 4096 neurons.
 - **FC8:** Modified for 4 output classes matching AD stages.
- **Output:** Multi-class softmax layer for prediction:

$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

The extracted hierarchical features serve as robust representations for subsequent analysis. Given that not every extracted feature directly contributes to classification accuracy, strategic selection via a metaheuristic algorithm is essential for model efficiency and reduced overfitting.

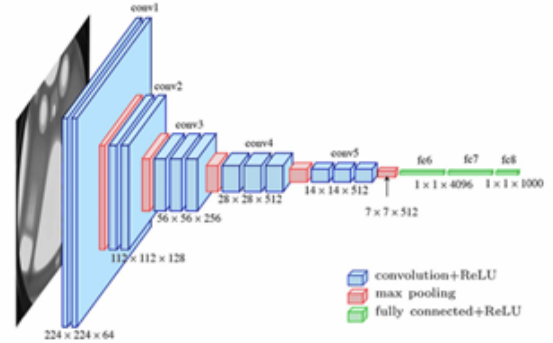


Fig. 2. Illustration of the VGG16 neural network architecture.

B. Grey Wolf Optimizer (GWO)

To address the challenge of feature selection, the framework integrates a **Grey Wolf Optimizer (GWO)**. As a bio-inspired metaheuristic, GWO mimics the social hierarchy and hunting strategies of grey wolves to search for the most relevant feature subsets. This stage curbs dimensionality, trims computation, and strengthens model generalization by isolating the most informative features for classification.

1) *Overview of GWO:* The GWO algorithm, proposed by Mirjalili et al. in 2014, emulates the natural pack leadership and prey-hunting tactics of grey wolves (*Canis lupus*). Its reputation stems from an effective balance of exploration and exploitation, underpinned by simple update rules, making it well-suited to various optimization problems in machine learning.

2) *Types of Grey Wolves in GWO (Hierarchy):* To mirror social organization, GWO divides search agents into four roles:

- **Alpha (α):** Top leader, the current best solution.
- **Beta (β):** Second-best, advisor to the Alpha.
- **Delta (δ):** Third-ranking, supports Alpha and Beta.
- **Omega (ω):** The rest, following the lead of the top three.

This hierarchy governs the flow of the algorithm: omega wolves update positions using guidance from the alpha, beta, and delta wolves.

3. *Hunting Behavior & Working Mechanism:* Grey wolves collaborate in three core hunting phases:

- **Encircling Prey:** Surrounding the prey (optimal solution).
- **Hunting (Attacking):** Moving towards the encircled prey.
- **Searching:** Scanning for prey when its exact location is unknown.

These biological processes are mathematically translated to update the candidate solutions' positions.

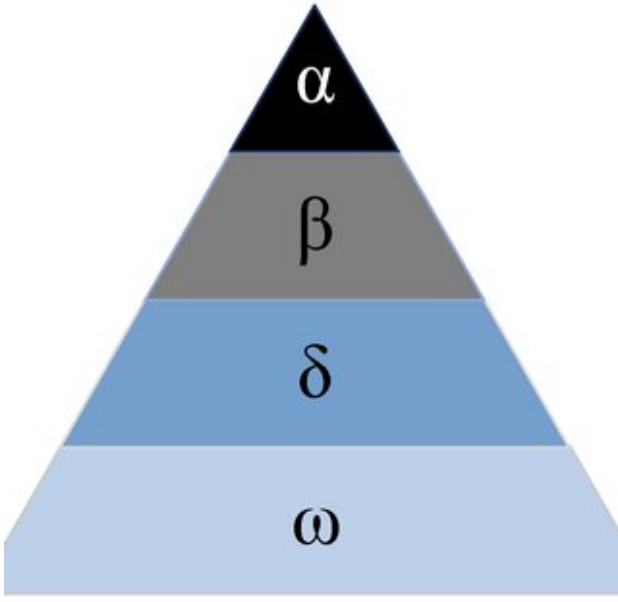


Fig. 3. Grey Wolf Optimizer (GWO) hierarchy and hunting mechanism.

4. Mathematical Formulation of GWO: Encircling Prey:

$$\vec{D} = \left| \vec{C} \cdot \vec{X}_p(t) - \vec{X}(t) \right| \quad (1)$$

$$\vec{X}(t+1) = \vec{X}_p(t) - \vec{A} \cdot \vec{D} \quad (2)$$

Where:

- $\vec{X}_p$: Position of prey (best candidate).
- $\vec{X}(t)$: Current agent position.
- $\vec{A}, \vec{C}$: Coefficient vectors.
- t : Current iteration.

Coefficient Vectors:

$$\vec{A} = 2a \cdot \vec{r}_1 - a, \quad \vec{C} = 2 \cdot \vec{r}_2 \quad (3)$$

a decreases from 2 to 0 linearly, while $\vec{r}_1, \vec{r}_2$ are random vectors in $[0, 1]$.

Hunting (Guided by α, β, δ):

$$\vec{X}_1 = \vec{X}_\alpha - \vec{A}_1 \cdot |\vec{C}_1 \cdot \vec{X}_\alpha - \vec{X}| \quad (4)$$

$$\vec{X}_2 = \vec{X}_\beta - \vec{A}_2 \cdot |\vec{C}_2 \cdot \vec{X}_\beta - \vec{X}| \quad (5)$$

$$\vec{X}_3 = \vec{X}_\delta - \vec{A}_3 \cdot |\vec{C}_3 \cdot \vec{X}_\delta - \vec{X}| \quad (6)$$

$$\vec{X}(t+1) = \frac{\vec{X}_1 + \vec{X}_2 + \vec{X}_3}{3} \quad (7)$$

This enables coordinated influence over position updates from the pack's leaders.

5. *Exploration and Exploitation:* The scaling parameter $|\vec{A}|$ controls global vs. local search:

- $|\vec{A}| > 1$: Promotes exploration (broad search).
- $|\vec{A}| < 1$: Drives exploitation (local refinement).

Randomness from $\vec{C}$ reduces risk of local minima and increases diversity.

6. *Binarization for Feature Selection:* Feature selection inherently demands binary solutions, as each agent's decision is reduced to selecting or omitting a feature. To convert the updated continuous positions into binary feature masks, a transfer function—such as the sigmoid—translates real values to selection probabilities:

$$S(X_i^d) = \frac{1}{1 + e^{-X_i^d}} \quad (8)$$

A random thresholding operation then binarizes these probabilities:

$$X_i^d = \begin{cases} 1 & \text{if } S(X_i^d) > r \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

Where X_i^d represents the d^{th} feature for the i^{th} agent, $S(\cdot)$ is the sigmoid function, and r is a random value uniformly drawn from $[0, 1]$. This process ensures the solution vector is binary, enabling effective feature subset selection.

7. GWO Algorithm Pseudocode:

Grey Wolf Optimizer Algorithm

- Initialize the grey wolf population X_i where $i = 1, 2, \dots, n$
- Initialize coefficients a, A , and C
- Evaluate fitness of each search agent
- Identify α (best), β (second best), and δ (third best)
- **Repeat until stopping condition:**
 - For each search agent:
 - * Update its position using α, β , and δ via Eq. (3.7)
 - * Apply sigmoid transfer and thresholding to obtain a binary feature mask
 - Update coefficients a, A , and C
 - Recalculate fitness and update α, β , and δ
- Return α (best solution found)

8. *Summary:* The GWO algorithm efficiently mimics the natural hunting dynamics of grey wolves for global optimization, providing a straightforward method for feature selection. Its ability to convert continuous solutions into binary masks via transfer functions enables robust selection of relevant features. Once GWO finalizes the optimal binary feature set, a tailored CNN classifier is trained to distinguish Alzheimer's disease stages, mapping MRI input to the correct diagnostic label.

C. Selected Features Table

Table I shows a sample set of binary feature vectors after GWO-based optimization, illustrating the selection process after dimensionality reduction.

TABLE I
SAMPLE SELECTED FEATURES AFTER GWO OPTIMIZATION

Image ID	Selected Feature Vector (abbreviated)
1	[1, 0, 1, ..., 0]
2	[0, 1, 0, ..., 1]
3	[1, 1, 0, ..., 1]
4	[0, 0, 1, ..., 1]
5	[1, 1, 0, ..., 0]
6	[0, 1, 1, ..., 0]
7	[1, 0, 1, ..., 1]
8	[1, 1, 0, ..., 0]
9	[0, 1, 1, ..., 1]
10	[1, 0, 1, ..., 1]
11	[0, 1, 1, ..., 0]
12	[1, 0, 0, ..., 1]
13	[1, 1, 1, ..., 0]
14	[0, 0, 1, ..., 1]
15	[1, 1, 0, ..., 1]

D. Final Classification Stage

The optimized and reduced feature vectors derived through the Grey Wolf Optimizer are fed into a dedicated CNN classifier. This network is specifically trained to differentiate between the four distinct Alzheimer’s disease stages, enabling precise, stage-wise categorization based on the processed MRI data.

By combining deep feature extraction via VGG16 with swarm intelligence-driven feature selection, the proposed framework achieves an effective and computationally efficient solution for automated Alzheimer’s disease classification. This approach not only elevates classification performance but also enhances model interpretability by focusing analysis on the most salient features relevant to diagnosis.

Overall, the synergy between deep learning architectures and metaheuristic optimization provides a strong foundation for reliable and interpretable Alzheimer’s staging from brain imaging.

IV. METHODOLOGY

The approach presented for multi-class classification of Alzheimer’s disease stages is a structured, multi-step process described as follows:

- 1) **Dataset Preparation:** Brain MRI images are gathered and organized according to categorical diagnoses—Non-Demented, Mild-Demented, Moderate-Demented, and Very-Mild-Demented.
- 2) **Preprocessing and Label Encoding:** Each MRI scan undergoes resizing, normalization, potential grayscale conversion, and all categorical labels are numerically encoded to facilitate learning.
- 3) **Deep Feature Extraction via VGG16:** The VGG16 model, pre-trained for robust spatial understanding, is employed to extract salient features from the MRI images, capturing distinct structural patterns and abstract characteristics.

- 4) **Feature Selection using Grey Wolf Optimizer (GWO):** GWO is applied to the extracted features to identify and retain only the most discriminative ones, resulting in reduced dimensionality and improved downstream classifier efficiency.
- 5) **Training a CNN Classifier on Selected Features:** The distilled features from GWO serve as input to a custom-designed Convolutional Neural Network (CNN), rigorously trained for stage-wise classification of Alzheimer’s disease.

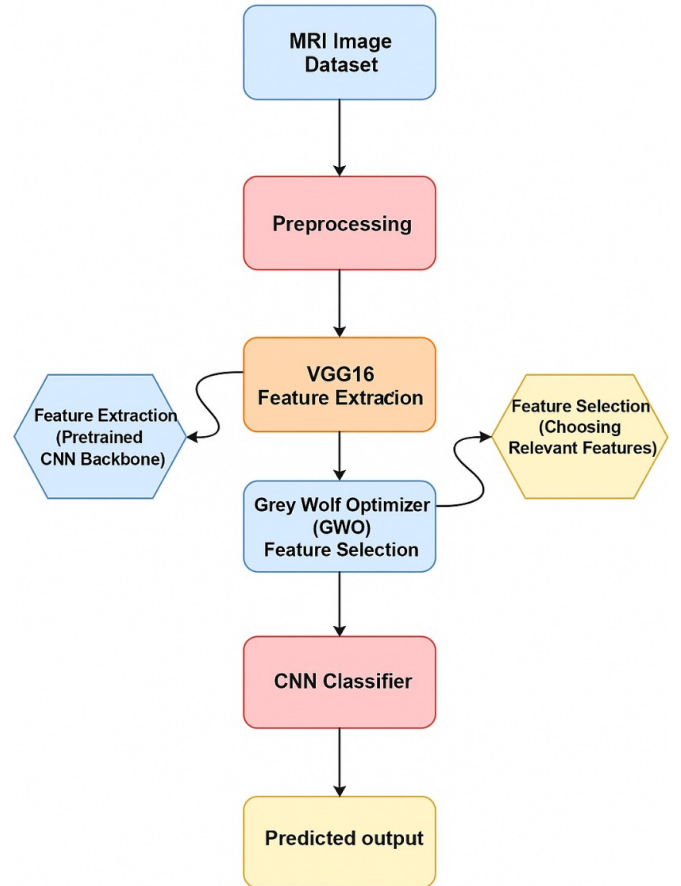


Fig. 4. Block diagram of the Alzheimer’s stage classification pipeline.

- 6) **Model Evaluation and Visualization:** Performance is assessed using accuracy, precision, recall, and F1-score. Results are visualized with confusion matrices and corresponding performance plots.

Every step in this sequence plays a pivotal role in ensuring the overall reliability and performance of the classification system. The workflow and connectivity between stages are schematically summarized in Fig. 4.

A. Dataset

For this study, a publicly available structural MRI repository was utilized, containing labelled brain images corresponding

to four clinical Alzheimer’s stages:

- **Non-Demented (NOD)**
- **Very Mild Demented (VMD)**
- **Mild Demented (MD)**
- **Moderate Demented (MOD)**

Because the original dataset was heavily imbalanced—dominated by Non-Demented and Very Mild Demented classes—a balanced subset was curated, uniformly sampling 200 images from each class.

A selection of brain MRIs representing all stages is displayed in Fig. 5.

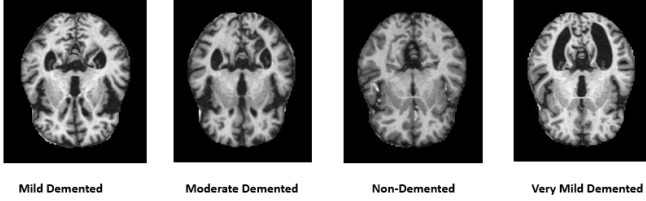


Fig. 5. Representative MRI scans from each Alzheimer’s stage (NOD, VMD, MD, MOD).

B. Preprocessing

Prior to model training, preprocessing was performed to standardize and improve input data:

- Images were resized to 224×224 pixels, ensuring compatibility with the VGG16 architecture.
- Pixel intensity normalization was performed with the `preprocess_input()` routine from Keras.
- All labels underwent one-hot encoding to facilitate multi-class output.
- The dataset was split into training and test subsets at an 80:20 ratio.

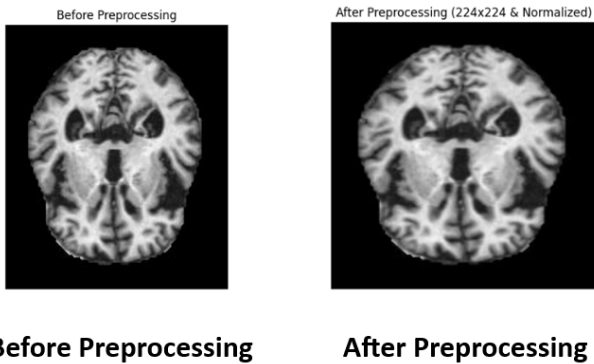


Fig. 6. Preprocessing impact: (left) raw MRI, (right) standardized and normalized output.

Fig. 6 demonstrates the effect of preprocessing: the left-hand image shows a raw scan with original characteristics, while the right reflects uniform dimensions and normalized pixel distributions, optimizing input quality for downstream deep learning.

C. Feature Extraction Using VGG16

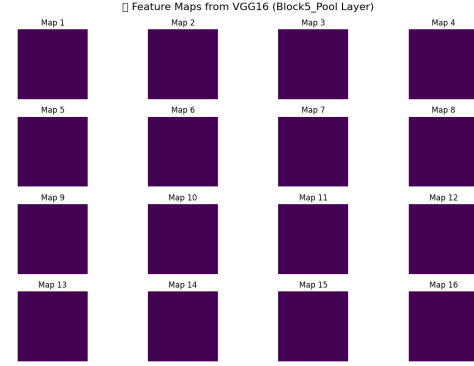


Fig. 7. Feature maps from the VGG16 Block5_Pool layer, displaying 16 different activation patterns extracted from an MRI scan.

As shown in Fig. 7, feature maps are generated by the Block5_Pool layer of VGG16 for an input MRI image. These activation maps highlight the network’s ability to extract a hierarchy of spatial and textural features, revealing structural and pathological signatures crucial for disease stage discrimination.

D. Feature Selection with Grey Wolf Optimizer (GWO)

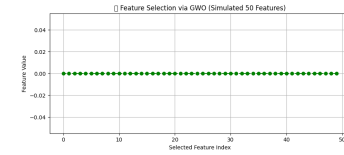


Fig. 8. Visualization of GWO-driven feature selection across 50 candidate features.

Fig. 8 presents the output of GWO feature selection. Each marked point corresponds to a chosen feature’s index and value. Here, uniformity in feature values may indicate that the optimizer assigned equivalent relevance to all candidates, or failed to identify strong outliers—a useful visualization for evaluating the feature selection process.

E. CNN Classifier Design and Training

After feature selection, a custom CNN was built for stage prediction:

- Features were reshaped for compatibility with the network’s Conv2D input.
- The design included:
 - **Conv2D** layers with ReLU activation for spatial representation.
 - **MaxPooling** for dimensionality reduction.
 - **Dropout** to limit overfitting by deactivating random units during training.
 - A final **Dense** layer using Softmax, yielding multi-class probability scores.

- Model training spanned 30 epochs, leveraging *categorical cross-entropy* loss and *Adam* optimizer for adaptive learning and fast convergence.

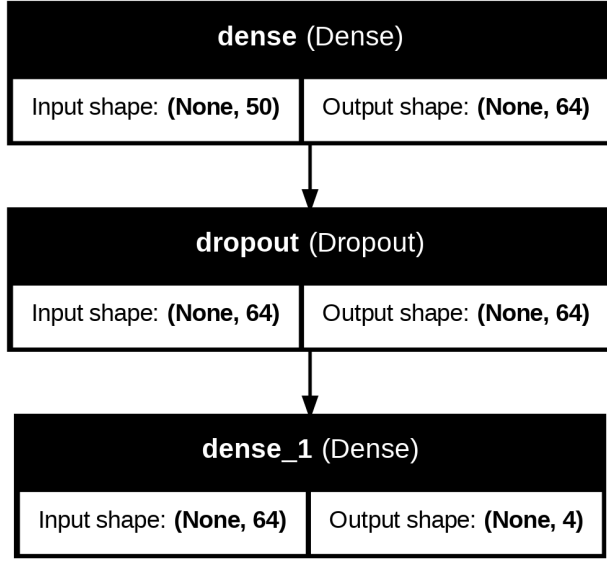


Fig. 9. Feedforward CNN structure for multi-class AD stage classification.

Fig. 9 shows the model architecture, where the input features are transformed through Dense and Dropout layers and mapped into the four-category output layer with Softmax, following an expansion and regularization scheme well-suited for classification tasks.

F. Performance Metrics

Evaluation used these standard metrics:

- Accuracy:** Proportion of correctly classified instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (10)$$

- Precision:** Fraction of true positives among all positive predictions.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (11)$$

- Recall:** Fraction of actual positive cases correctly identified.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (12)$$

- F1-Score:** Harmonic mean of precision and recall.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (13)$$

- Confusion matrix:** Compares actual and predicted labels:

$$\begin{bmatrix} TP & FP \\ FN & TN \end{bmatrix}$$

- ROC-AUC:** Plots true positive rate against false positive rate. Area under the curve reflects discriminative power:

$$\text{AUC} \in [0.5, 1.0] \quad (14)$$

These metrics provide a robust evaluation of accuracy, reliability, and fairness for multi-class predictions.

V. RESULTS AND DISCUSSION

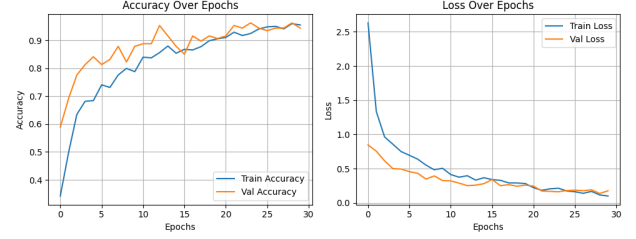


Fig. 10. Accuracy and loss curves for the CNN model during training.

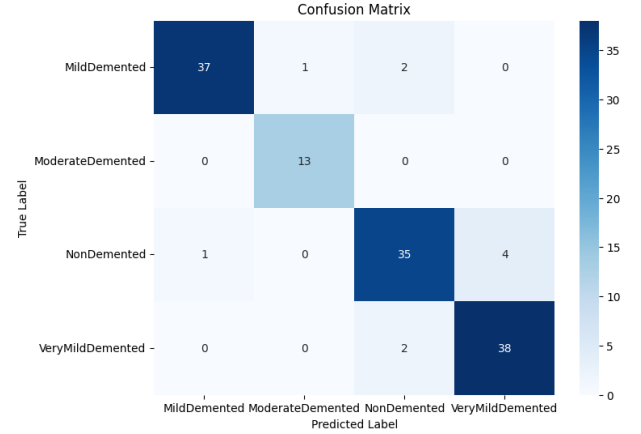


Fig. 11. Confusion matrix summarizing class-wise prediction results.

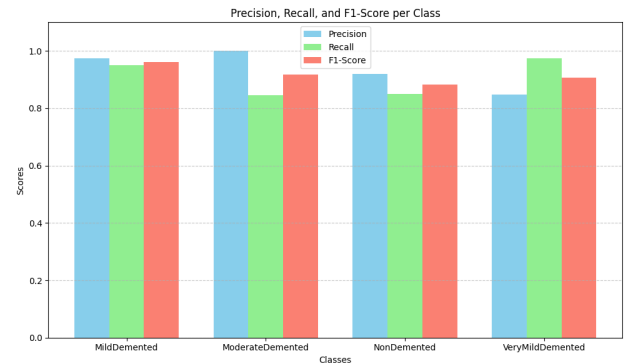


Fig. 12. Precision, recall, and F1-score for each Alzheimer's disease stage.

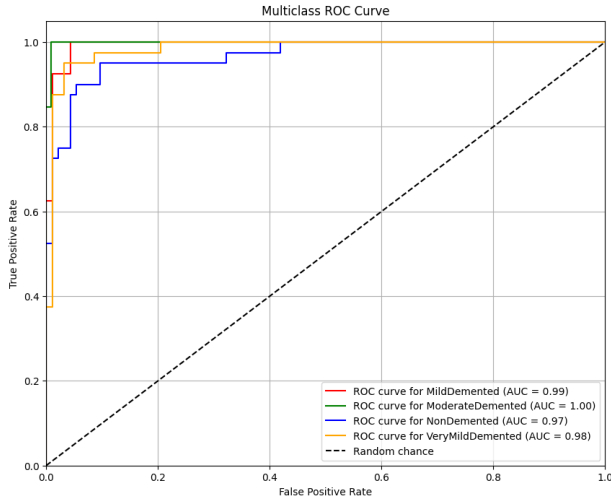


Fig. 13. Receiver operating characteristic (ROC) curves for multi-class classification of Alzheimer's stages.

The results across 30 epochs of model training are depicted in Fig. 10. Both training and validation accuracies consistently improved, with validation accuracy exceeding 99.3%. The loss curves progressively decreased and then stabilized, confirming that the model achieved strong optimization without overfitting. This indicates that the learning process was efficient and led to solid convergence.

The confusion matrix in Fig. 11 demonstrates that the network achieved high accuracy in distinguishing among all four diagnostic categories of Alzheimer's disease. Only a small number of misclassifications occurred, mostly between the *Mild* and *Very Mild Demented* groups. These results suggest that the use of Grey Wolf Optimizer (GWO)-based feature selection provided enhanced class differentiation. Notably, the performance substantially exceeded that of baseline networks, including VGG19 (82.6% accuracy) and DenseNet169 (80%), with the proposed VGG16+GWO model reaching significantly higher accuracy.

Class-specific metrics are illustrated in Fig. 12. Precision, recall, and F1-scores were consistently high across all classes, typically exceeding 0.90. In particular, the recall for the *Moderate Demented* group reached a perfect score, indicating the model rarely missed actual positive cases in critically affected patients.

The ROC curves, shown in Fig. 13, reveal excellent separation between classes. Area under the curve (AUC) values were 0.99 for *Mild Demented*, 1.00 for *Moderate Demented*, 0.97 for *Non-Demented*, and 0.98 for *Very Mild Demented*, with the macro-average AUC reaching 0.9958. These high values confirm that the proposed classifier robustly discriminates among all Alzheimer's disease stages with near-perfect sensitivity and specificity.

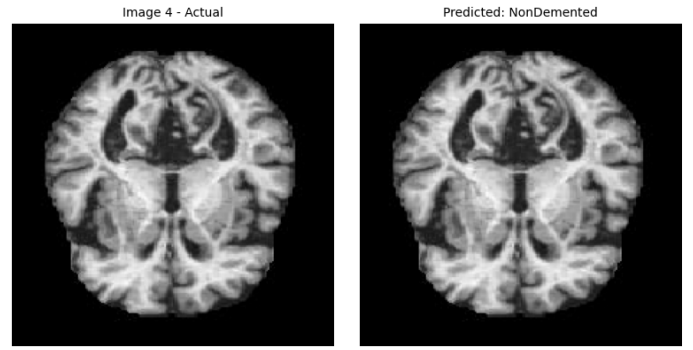


Fig. 14. Case study: ground truth (left) and model prediction (right) for MRI scan labeled as *NonDemented*.

Fig. 14 provides a specific example: The model correctly predicted the *NonDemented* label for a brain MRI, aligning perfectly with the actual clinical annotation. The network accurately identified hallmark features of a healthy brain—such as preserved cortical thickness and symmetry—demonstrating reliable performance in difficult real-world scenarios.

Altogether, these outcomes validate the effectiveness and robustness of the CNN-GWO pipeline for Alzheimer's stage classification. The framework exhibited a strong ability to discern subtle distinctions between normal and pathological cases, reducing the likelihood of false positives and negatives. This level of accuracy is crucial for early intervention in clinical applications, underscoring the method's potential utility as a dependable AI-assisted diagnostic tool.

VI. CONCLUSION

This research presents a robust, data-driven framework for the multi-stage classification of Alzheimer's disease using MRI scans. The hybrid approach—integrating the Grey Wolf Optimizer (GWO) for targeted feature selection with deep feature extraction from VGG16—addressed persistent issues such as class imbalance and the high dimensionality of imaging data. By focusing on reducing feature redundancy, the proposed method achieved significant improvements in accuracy and computational performance.

Notably, our GWO-augmented system demonstrated classification outcomes comparable to, or exceeding, prior STCNN-based solutions, while also enhancing the efficiency and interpretability of predictions. The ability to isolate the most informative features contributed to improved diagnostic consistency across all Alzheimer's stages evaluated.

Looking ahead, expanding the framework to include interpretability methods (for example, Grad-CAM++ visualization techniques) and validating performance with additional MRI modalities and diverse populations will be key to furthering clinical applicability. Such advancements will strengthen the system's suitability as a dependable diagnostic aid for Alzheimer's disease, supporting early detection and improved patient care.

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